

# Uddesh Santosh Kumar Singh

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## Education

University of Texas at Dallas, MS in Computer Science GPA: 4.0 | Jan 2026  
National Institute of Technology, Allahabad, B.TECH in Computer Science & Engineering GPA: 3.3 | Jun 2020

## Skills

**Programming Languages:** Python, Go (Golang), Java, C++, C, SQL, JavaScript, TypeScript, Rust, Bash/Shell Scripting  
**Distributed Systems & Infrastructure:** GCP (GKE, Cloud Run, Pub/Sub), Kubernetes (K8s), Docker, Apache Kafka, gRPC, Redis, Terraform  
**Backend & Web Development:** FastAPI, Spring Boot, .NET Core, React, Node.js, RESTful APIs, GraphQL, Microservices, PostgreSQL, MySQL  
**AI & Machine Learning:** Generative AI, RAG, Information Retrieval, Vector DBs (FAISS), LangChain, TensorFlow, PyTorch, Vertex AI  
**Developer Tools:** Git, GitHub Actions, CI/CD, Jenkins, Linux/Unix, PyTest, JUnit, Selenium, Postman, System Design

## Experience

**Software Development Intern**, Paycom – Irving, TX May 2025 – Aug 2025  
• Accomplished **120x processing efficiency** for high-volume data extraction by architecting a cloud-native **C#** and **.NET** service.  
• Delivered **\$250k in annual savings** by deploying automated infrastructure pipelines leveraging **AWS**, **Docker**, and **GCP** services.  
• Eliminated production regressions across **2,000+ lines of code** by integrating **xUnit** into **CI/CD** and **GitHub Actions**.

**Technical Product Owner (Senior Associate)**, American Express – Gurugram, HR Jan 2024 – July 2024  
• Increased high-confidence alert accuracy by **40%** by scaling real-time monitoring systems using **PostgreSQL** and distributed data analytics.

**Software Engineer II (Senior Associate)**, American Express – Gurugram, HR Nov 2022 – Jan 2024  
• Prevented **\$6M in annual fines** by designing a high-concurrency **Golang** distributed system using **gRPC** and **GORM**.  
• Reduced **API latency by 60%** by migrating legacy monolithic services to **Python (FastAPI)** microservices deployed on **GCP** Kubernetes.  
• Improved data discovery by **30%** by engineering internal search systems using **Vector Embeddings** and **Information Retrieval** pipelines.  
• Optimized system throughput for **12k+ DAU** by implementing multi-level **Redis caching** and parallel processing in **Linux** environments.

**Software Engineer I (Programmer Analyst)**, American Express – Gurugram, HR Sept 2020 – Oct 2022  
• Accelerated engineer onboarding by **400%** by developing a **Python (Flask)** and **React** metadata health-checker service for **MySQL**.  
• Boosted internal tool adoption by **25%** by modernizing legacy portals into responsive **Single Page Applications** using **JavaScript**.  
• Guaranteed production fault tolerance by enforcing **100% code coverage** through rigorous **PyTest (TDD)** and **Unix** shell scripting.

**Business Technology Analyst Intern**, Deloitte USI – Hyderabad, TG May 2019 – July 2019  
• Maintained **100% feature parity** during the migration of a mission-critical warehousing application to **Android** using **Java**.  
• Eliminated manual regression testing bottlenecks by integrating **183+ automated Selenium test cases** into the **CI/CD** pipeline.

## Projects

**Watchparty** github.com/uddeshsingh/WatchParty  
• Achieved **~50ms stream latency** for **10k+ concurrent users** by developing a high-concurrency engine in **Golang** and **WebSockets**.  
• Synchronized global playback states across thousands of clients by architecting a **distributed messaging system** using **Apache Kafka**.  
• Reduced server CPU load by **30%** by implementing optimized **Goroutines** and **Unix** socket handling for real-time data delivery.

**AutoVid** github.com/uddeshsingh/YTBot  
• Reduced production effort by **90%** by architecting a fault-tolerant **GenAI** platform using **FastAPI** and **Kubernetes** rendering agents.  
• Scaled system throughput for parallel video generation by streamlining asynchronous task execution with **Python**, **Celery**, and **Docker**.

**SIA** github.com/uddeshsingh/SIA  
• Cut inference latency by **40%** by developing an **AI financial app** utilizing **Next.js** and **RAG** data ingestion pipelines.  
• Increased context retrieval accuracy by **35%** by fine-tuning **FAISS VectorDBs** and **LangChain** for semantic financial **Information Retrieval**.

## Achievements

Code-a-thon Winner, 2025, Paycom  
Certified Software Engineer and Cloud Engineer, Google Inc  
Associate of the Quarter, Q2 2023 & Q1 2022, American Express